Cluster Before You Train

Fingerprint-Based Concept Discovery for Federated Learning Under Recurrent Drift

April 2026

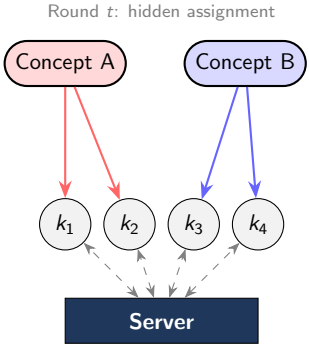
Outline

Problem Setting: Federated Learning Under Concept Drift

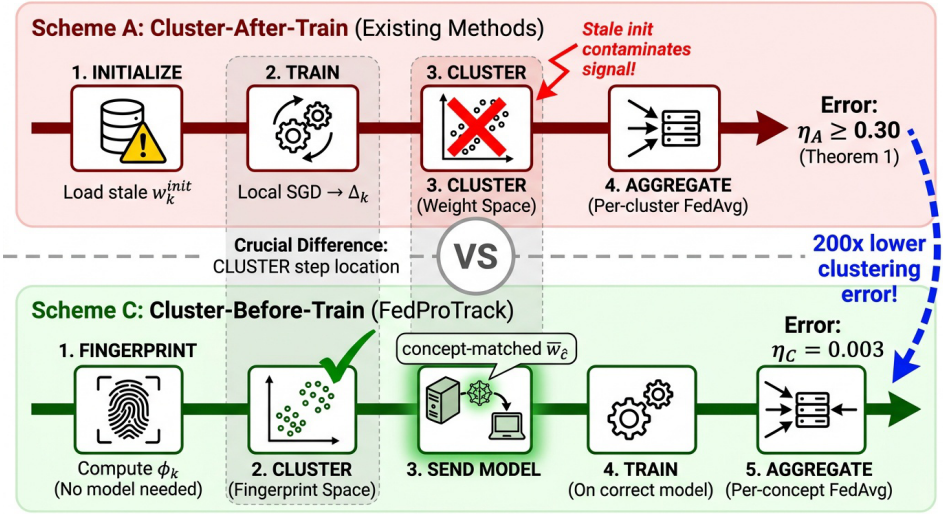
- K clients, T rounds, C latent **concepts**
- Each round, client k draws data from concept $c(k,t) \in [C]$
- Concepts **recur**: client may revisit concept a after visiting b
- Server cannot see raw data — only aggregated signals

Goal: infer latent concept identity $\rightarrow$ group clients $\rightarrow$ per-concept federated aggregation

Oracle (upper bound): knows true $c(k,t)$ every round



The Core Design Choice: When to Cluster?

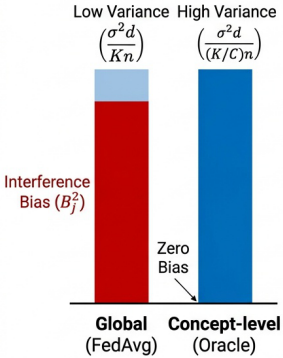


Thm 1: $\eta_A \geq \eta_A^{(\text{clean})} + \kappa \cdot \mathbb{V}(w^{(\text{init})})$ $\Rightarrow$ Scheme A: $\eta \geq 0.30$ vs Scheme C: $\eta = 0.003$

When Does Concept-Level Aggregation Help?

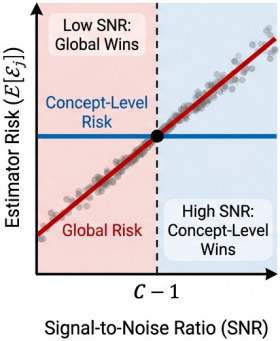
The Bias-Variance Tradeoff

Risk formulas for Global vs. Concept-Level aggregation



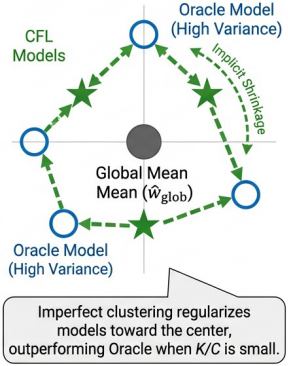
The Sharp Crossover

Analytical crossover condition ($\mathrm{SNR} > C - 1$) validated by synthetic experiments



Implicit Shrinkage Mechanism

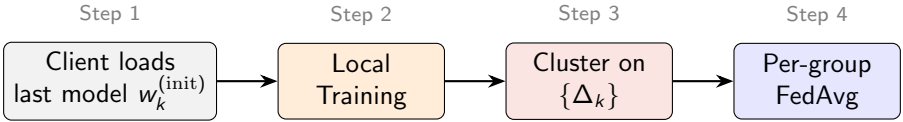
Conceptually: Imperfect clustering on the model weight space



Left: Bias-variance tradeoff **Center**: Sharp crossover at $\mathrm{SNR} = C - 1$ **Right**: Imperfect clustering = implicit shrinkage

Existing Methods: Scheme A (Cluster-After-Train)

Nearly all clustered FL methods follow the same protocol:



Examples: IFCA, CFL, FedEM, FedRC, FedGWC, gradient-based partitioning ...

The Hidden Assumption

The weight update Δ_k faithfully reflects the client's *current* concept identity.

But under concept recurrence, this assumption breaks.

The Stale-Initialisation Problem

When client switches from concept $a \to b$, it loads a model last trained on a:

$$\Delta_k = \underbrace{-\alpha \nabla L_b(\bar{w}_b)}_{\text{true concept-}b\text{ signal}} + \underbrace{\alpha H_k \left(w_k^{(\text{init})} - \bar{w}_b^{(t-1)} \right)}_{\text{stale-init contamination}} + O(\| \cdot \|^2)$$

- The contamination term has **nothing to do** with concept b
- It is proportional to $\| w_k^{(\text{init})} - \bar{w}_b \|$ — the "mismatch"
- Any clustering algorithm operating on Δ_k is **structurally misled**

$\Rightarrow$ **This is not an algorithm problem — it is a protocol structure problem.**

Theorem: Stale-Initialisation Amplification

Theorem (Main Result — Unconditional)

Under standard regularity conditions (L-smooth, μ-strongly convex):

$$\boxed{\eta_A \;\geq\; \eta_A^{(\mathrm{clean})} \;+\; \kappa \cdot \mathbb{V}(w^{(\mathrm{init})})}$$

where $\mathbb{V}(w^{(\mathrm{init})}) > 0$ **whenever concepts recur**.

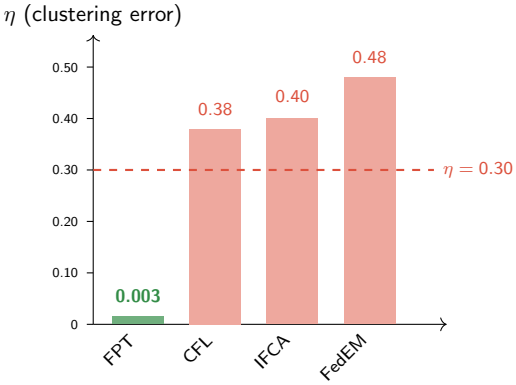
Three key implications:

1. **Unconditional**: holds under standard assumptions, no extra conditions needed
2. $\mathbb{V}(w^{(\mathrm{init})}) > 0$ is **structural** under concept recurrence — **cannot be removed**
3. No algorithm improvement within Scheme A can close this gap

From clustering error to excess risk (Corollary 1):

Combining with the implicit-shrinkage characterisation gives excess-risk gap $\geq (\eta_A - \eta_C) \cdot \frac{C}{C-1} \cdot \bar{B}^2 > 0$

Empirical Evidence: $\eta \geq 0.30$ For All Scheme A Methods



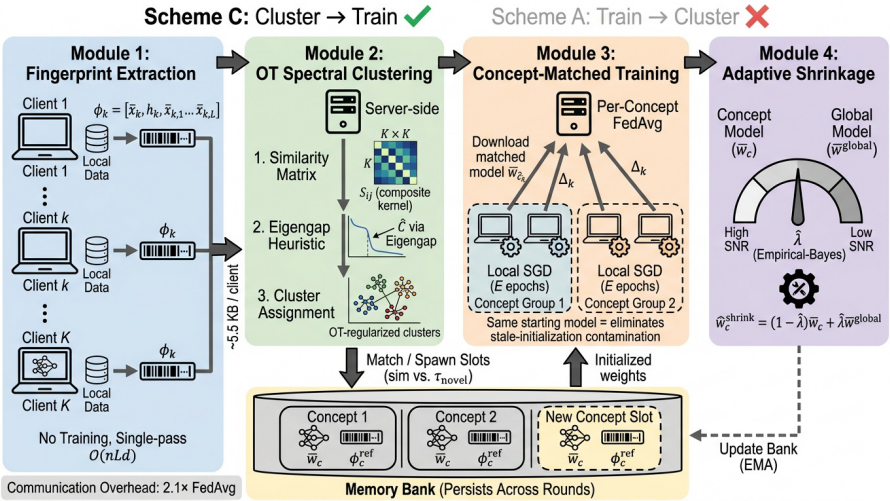
- On CIFAR-100, **all** Scheme A methods: $\eta \geq 0.38$
- FPT achieves $\eta = 0.003$ — **two orders of magnitude lower**
- This gap is *structural*, not algorithmic

Even with Oracle initialisation (removing $\mathbb{V}$): $\eta_A^{(\text{clean})} = 0.40$ vs $\eta_C = 0.002$

$\Rightarrow$ Fingerprints have inherent d/L SNR advantage

(Lemma 2: $d = 128$, $L = 10 \Rightarrow 12.8\times$)

FedProTrack: Overview



Scheme C: Fingerprint → OT Spectral Clustering → Concept-Matched Training → Adaptive Shrinkage (2.1× FedAvg)

Module 1: Fingerprint Extraction

Each client computes a **data fingerprint** without any model training:

$$\phi_k = \Big[\underbrace{\bar{x}_k}_{\text{global mean}}, \underbrace{h_k}_{\text{label dist.}}, \underbrace{\bar{x}_{k,1}, \ldots, \bar{x}_{k,L}}_{\text{class-cond. means}}\Big]$$

Why this design?

- Class-conditional means = **sufficient statistics** for concept identity
- Single-pass $O(nLd)$, no gradient needed
- d=128, L=10 $\Rightarrow$ **5.5 KB** per client

SNR advantage (Lemma 2):

$$\frac{\mathrm{SNR}_{\mathrm{fp}}}{\mathrm{SNR}_{\mathrm{wt}}} = \frac{d}{L} = \frac{128}{10} = 12.8\times$$

Fingerprints encode concept signal as *first-order* moments; weights bury it in *quadratic* forms ($X^\top X$).

Module 2: OT Spectral Clustering — Step 1: Similarity Matrix

Given K fingerprints $\{\phi_1, \ldots, \phi_K\}$, the server computes:

$$S_{ij} = \underbrace{0.25 \cdot \mathrm{sim}_{\mathrm{feat}}(\phi_i, \phi_j)}_{\text{global-mean Gaussian kernel}} + \underbrace{0.30 \cdot \mathrm{sim}_{\mathrm{label}}(h_i, h_j)}_{\text{label-histogram cosine}} + \underbrace{0.45 \cdot \mathrm{sim}_{\mathrm{cc}}(\phi_i, \phi_j)}_{\text{class-conditional Gaussian kernel}}$$

- Weights fixed on held-out synthetic data, **not tuned** on evaluation
- $\mathrm{sim}_{\mathrm{cc}}$ dominates: class-conditional structure carries the most concept signal
- **Robust**: accuracy varies ≤ 0.2pp across 4 weight configurations

Theoretical connection: for Gaussian classes with shared covariance, Euclidean distance on class-conditional means $\approx$ **Wasserstein-2**: $W_2^2(P_a, P_b) = \sum_c w_c \|\mu_a^{(c)} - \mu_b^{(c)}\|^2$

Module 2: OT Spectral Clustering — Step 2: Eigengap

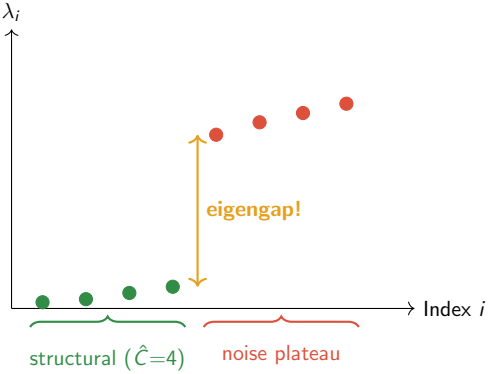
Automatic concept-count selection via normalised Laplacian:

1. Construct $L_{\text{sym}} = I - D^{-1/2} S D^{-1/2}$
2. Compute eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_K$
3. Find the largest gap:

$$\hat{C} = \arg\max_{c \in [2, K/2]} (\lambda_{c+1} - \lambda_c)$$

Intuition: C well-separated clusters $\Rightarrow$ C eigenvalues ≈ 0, then a **jump**

Result: Eigengap correctly recovers $\hat{C}$=4 in >95% of rounds. Fixing $\hat{C}$=C costs only 0.6pp.



Module 2: OT Spectral Clustering — Step 3: OT-Regularised Assignment

Standard spectral clustering: top-$\hat{C}$ eigenvectors $\rightarrow$ row-normalise $\rightarrow$ k-means.

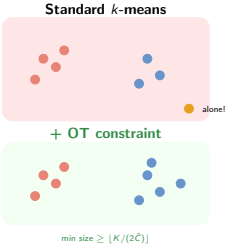
Problem: k-means may produce empty/tiny clusters — a 1-client group has no aggregation benefit.

OT regularisation replaces k-means:

$$\min_{\pi} \sum_{i,c} \pi_{ic} \|v_i - \mu_c\|^2$$

$$\text{s.t.} \quad \sum_{c} \pi_{ic} = 1 \quad \forall i$$

$$\sum_{i} \pi_{ic} \geq \lfloor K/(2\hat{C}) \rfloor \quad \forall c \quad \text{(min size)}$$



Module 3: Memory Bank & Concept Matching

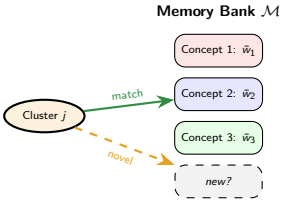
Memory bank $\mathcal{M} = \{(\bar{w}_c, \phi_c^{\text{ref}})\}$ stores model + fingerprint per concept.

Matching: each new cluster centroid vs stored concepts:

- $\text{sim} \geq \tau_{\text{novel}}$: re-identify $\rightarrow$ load stored model
- $\text{sim} < \tau_{\text{novel}}$: new concept $\rightarrow$ fresh slot

τ_{novel}: 95th percentile of pairwise sim distribution

Key: dormant concepts are *recalled*, not re-learned.



Module 4: Concept-Matched Training & Adaptive Shrinkage

Concept-Matched Training

Client k in concept $\hat{c}_k$ receives $\bar{w}_{\hat{c}_k}^{(t-1)}$ from memory:

$$\Delta_k^{(t)} = w_k^{(t)} - \bar{w}_{\hat{c}_k}^{(t-1)}$$

$$\bar{w}_c^{(t)} = \bar{w}_c^{(t-1)} + \frac{1}{|\mathcal{C}_c|} \sum_{k \in \mathcal{C}_c} \Delta_k^{(t)}$$

All clients in $\mathcal{C}_c$ start from the **same** model $\Rightarrow$ **pure concept signal**, no stale-init.

Adaptive Shrinkage

$$\hat{w}_c^{\text{shrink}} = (1-\hat{\lambda})\bar{w}_c^{(t)} + \hat{\lambda}\,\bar{w}^{(t)}$$

$$\hat{\lambda} = \frac{\hat{\sigma}^2/((K/\hat{C})n)}{\hat{\sigma}^2/((K/\hat{C})n) + \hat{\sigma}_B^2}$$

- **High SNR**: $\hat{\lambda} \to 0$, fully concept-specific
- **Low SNR**: $\hat{\lambda} \to 1$, degrades to FedAvg

Safety net: even if clustering is noisy or SNR is low, FPT never hurts much.

Experimental Setup

4 Datasets:

- **CIFAR-100** (flagship, 5 configs)
- **Fashion-MNIST**
- **fMoW** (natural temporal drift)
- **CIFAR-10** (negative control)

K=20, T=100, C=4, recurrence period ρ=25

Frozen ResNet-18 features, PCA to d=128
Linear classifier (1290 params)
Identical budgets: 5 epochs, lr=0.01

26 Baselines from 7 Families:

- Cluster-after-train: IFCA, CFL, FedEM, FedRC
- Cluster-late: HCFL, FedGWC
- Drift-aware: FedDrift, Flash, FedCCFA
- Personalized: FedProx, APFL, Ditto, pFedMe
- Core: FedAvg, Oracle, LocalOnly
- Misc: FLUX, SCAFFOLD, ...

Each baseline uses **published defaults**; grid-search on held-out config.

Main Results

	Accuracy (↑)				Clust. Err.
Method	CIFAR-100	F-MNIST	fMoW	CIFAR-10	η (C-100)
Oracle	.673	.974	.599	.940	.000
FedAvg	.631	.912	.506	.917	—
CFL	.623	.922	.507	.916	.38
IFCA	.599	.821	.502	.713	.40
FedEM	.523	.866	.416	.857	.48
APFL	.640	.924	.515	.905	—
Flash	.635	.920	.508	.917	—
FPT (Ours)	**.670**	**.973**	**.584**	.907	**.003**
Oracle gap (pp)	4.7	6.1	9.3	2.3	
Gap closed	**93%**	**98%**	**84%**	−43%	

FPT closes 84–98% of Oracle gap (gap $>$ 4pp) — All Scheme A: $\eta \geq 0.38$, below FedAvg — CIFAR-10 (neg. ctrl): FPT degrades gracefully

Ablation: Every Module Matters

Variant	Accuracy	η
FPT (full)	$.716 \pm .031$	$.003 \pm .001$
– Concept discovery	$.684 \pm .028$	—
– Adaptive shrinkage	$.704 \pm .033$	$.003 \pm .001$
– Fingerprint $\rightarrow$ weights (Scheme A)	$.688 \pm .035$	$.39 \pm .04$
– Eigengap (fixed $\hat{C}=C$)	$.710 \pm .030$	$.004 \pm .001$

Key row: fingerprint $\rightarrow$ weight-space clustering

η: $0.003 \rightarrow 0.39$, Acc: $0.716 \rightarrow 0.688$

$\Rightarrow$ Isolates the **protocol ordering effect**

- – concept discovery: -3.2pp
- – shrinkage: -1.2pp
- – eigengap: -0.6pp (reliable)

Additional Analysis

K-Scaling

- CFL matches Oracle at K=4 (implicit shrinkage helps when K/C is small)
- CFL falls behind by 3.7pp at K=20
- FPT: $\eta \leq 0.01$ at **all** K values
- Confirms Corollary 1 (implicit shrinkage)

Communication Cost

- FPT: 2.1× FedAvg per round
- Fingerprint: 5.5 KB upload
- Server clustering: $<$ 1ms (K=20)
- Delivers +3.0–6.9pp over best 1× baseline

Boundary Conditions

- **No recurrence** (ρ=15, T=30): FPT $\approx$ FedAvg within -0.1pp ($\hat{\lambda} \to 1$, shrinkage dominates)
- **End-to-end CNN** (EMNIST): FPT achieves 89–90% (mid-pack) Feature drift breaks fingerprints

Weight Sensitivity

- Kernel weights $(\alpha_1, \alpha_2, \alpha_3)$: 4 configurations tested
- Accuracy varies ≤ 0.2pp
- η_C unchanged

Limitations

1. **Frozen-backbone assumption** — FPT relies on stable feature representations; end-to-end training breaks cross-round fingerprint comparability (EMNIST: mid-pack)
2. **Gaussian linear model for theory** — Theorem 1 proven under L-smooth, strongly-convex losses; non-convex extension is future work
3. **Scale** — Experiments use $K \leq 40$; spectral clustering is $O(K^3)$, scaling to $K > 1000$ needs Nyström approximation
4. **Privacy** — Fingerprints are aggregate statistics (not raw data), but formal DP analysis is beyond scope

Takeaways

1. **"When to cluster" is a critical design dimension**
   Scheme A suffers from **unconditional** stale-init amplification — protocol-level, not algorithmic
2. **Data fingerprints $>$ weight updates for concept discovery**
   d/L SNR advantage: $\eta_C = 0.003$ vs $\eta_A \geq 0.38$ (two orders of magnitude)
3. **FedProTrack: a complete Scheme C instantiation**
   OT spectral clustering + concept memory + adaptive shrinkage = **84–98% Oracle gap closed** across 4 datasets, 26 baselines (2.1$\times$ FedAvg communication)
4. **Adaptive shrinkage = safety net**
   Graceful degradation to FedAvg when concept structure is weak